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Inventor(s)

Acres; John F. et al.

TICKETS FOR TIPPING OR HAVING ENHANCED VALUE

Abstract

Systems and methods are directed to distributed systems that include networked electronic gaming machines. Before, after, or during play, a player may one or more credits or other value may be designated by a player at the electronic gaming machine. The one or more credits may be associated with and/or otherwise transferred to a tip ticket. The one or more credits may be associated with and/or otherwise transferred to an enhanced value ticket.

Inventors: Acres; John F. (Las Vegas, NV), Acres; Noah (Las Vegas, NV)

Applicant: Acres Technology (Las Vegas, NV)

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Background/Summary

FIELD OF THE INVENTION

[0001] This disclosure relates generally to gaming systems, and more particularly to systems that connect electronic gaming machines on a network and that provide services to the gaming machines as well as management functions.

BACKGROUND

[0002] Electronic gaming machines, such as slot machines or video poker machines, employ at least one computer processor that generates a random outcome for each game and controls how the outcome is revealed to a player, e.g., using a stepper motor to spin reels on a slot machine or showing cards on a video display of a poker machine. There are also a variety of associated devices that enable wagering and play. For example, wager buttons for selecting the amount of a wager, a game initiation button, a ticket reader, bill validators, speakers, lights, and key pads, combinations, thereof, and the like.

[0003] Gaming machines may operate on a stand-alone basis or they may be connected to a network of gaming machines (e.g., networked gaming machines). Networked gaming machines may have a variety of services provided to them via a remote server (e.g., in a secure room or from a secure server on the gaming floor or in a remote location). These may include progressive jackpots, player tracking, electronic funds transfers to or from the gaming machine, cashless ticket transactions (known as ticket-in, ticket out or TITO) and dispatch of casino employees to a gaming machine, among others. In addition, the operator of the networked gaming machines can collect data from the network at a remote server, also in a secure room, including accounting information that can be compiled into various reports. This enables the operator to collect information such as amounts wagered and paid in awards, to account for cashless ticket transactions, and to track the amount wagered by each player enrolled in a player tracking club.

[0004] There are a number of technical problems associated with current systems on such networks. Current systems require a slot machine interface board (SMIB) to be installed in each gaming machine. The SMIB interfaces between the gaming machine and the network and requires its own power supply. The SMIB board is bulky and combined with its power supply takes up a lot of space in the gaming machine cabinet. The SMIB may be a multi-protocol interface that monitors communications between the network and at least one gaming machine. It can communicate, in a variety of protocols if necessary, over the network with hosts, i.e., remote servers, that provide the services and accounting features mentioned above.

[0005] Each SMIB is connected to a bank controller, which accumulates information from a plurality of the SMIBs and sends it to the servers in the secure room. Each SMIB also receives information from the servers, such as approvals for electronic fund transfers to and from the gaming machine and cashless ticket transactions, which it then relays to the gaming machine. Like the SMIB, each bank controller has its own power supply. There may be numerous bank controllers in a large casino.

[0006] In addition, the current system requires network switches, also each with its own power supply, to direct traffic on the network. As a result of these SMIBs, bank controllers, and network switches, and their respective power supplies, current systems are difficult to install and maintain.

[0007] Still another problem with the current systems is that they are proprietary so interfacing with third-party vendors is difficult. For example, if a casino wishes to use a different vendor from the one that provided the integrated suite (accounting, cashless tickets, loyalty, and audit functions), (e.g., to provide a progressive bonus), interfacing another company's product with the network is not easy. Interoperability among systems provided by different manufacturers is difficult, expensive, and often impossible.

[0008] One reason for these difficulties relates to the technology used in current systems, which are mostly provided by several large companies that have been using the same technology that was developed in the mid-1990s. A third-party vendor needs access to the network data to provide services such as bonuses, progressive jackpots, dispatch, or any other service. For security reasons, this is typically provided via an application programming interface (API) that accesses a read-only database, which stores data from the network. The system provider controls what data is available in the API database as well as the architecture of the API. If the systems provider changes either of these, a third-party vendor's system that uses the API may be adversely affected.

[0009] Many transactions, such as bonus or progressive jackpots awards, must be quickly executed. Data is not able to move quickly enough from the gaming machine, through the SMIB, its associated bank controller, and into the API database for processing by the vendor's system, which then must send a command, such as a jackpot pay, back through the various network components to the gaming machine in a timely fashion.

[0010] Reliability is a problem with the prior art systems. If a network connection is broken, the gaming machines downstream of the break are not able to function because ticket validation and electronic funds transfers, which are provided by the network, are not available. What is more, if a server crashes, the entire floor is down, i.e., the gaming machines are unplayable.

[0011] The SMIB typically includes a large capacitor to temporarily maintain voltage after a power failure and some ferromagnetic RAM that can store data without power to preserve the accounting and transaction data, which can then be accessed when the power is restored. But power backup and management is problematic because of all the power supplies—for the SMIBs, the bank controllers, and the network switches—that must all be dealt with in a coordinated manner when power is lost.

[0012] Because the hardware and its functionality may already be approved by gaming regulators, third-party developers of new software services for EGMs may gain ready approval for features provided by the new services. Put differently, a third-party developer need only gain approval of its software and the services it provides, rather than having to gain regulatory approval for hardware or interfaces with the system. This will enable small or even individual developers to create bonus games or other new services without the need to develop hardware and obtain extensive regulatory approval.

[0013] The present system addresses all these technical problems in a highly secure environment and provides additional benefits as will become apparent in the following description and related drawings.

[0014] The present system is well suited for implementing improvements in cashless ticket transactions. For example, it is known in the art that a player can print a ticket that has fewer than all of the credits on a credit meter, i.e., the player can allocate a portion of the credits on the credit meter to a ticket and leave the balance on the credit meter for further play by the player. The ticket can then be used to tip a casino employee who can either apply the ticket credits to the credit meter of a gaming machine and play games or cash out the ticket. It would be desirable to electronically implement for tip tickets. Some casinos pool all tips and divide the pooled tips among employees. Others allow each employee to keep their tips. The present improvement facilitates improved tip tickets in both circumstances.

[0015] In another example of an improved cashless ticket transaction, it would be desirable to motivate players in various ways by providing them with enhanced value tickets. For example, a player could receive a communication, e.g., via a display on a gaming device, that if they cash out now, they will receive a ticket that is worth more than the credits on the meter but only for play at a later date. Or the player could be informed that if they cash out now, they will receive a ticket with an enhanced amount of credits but only for play on a specific type of gaming machine or perhaps a sports wager. Another form of ticket enhancement is to provide a ticket that multiplies wins, e.g., doubles a jackpot on a game paid for with the enhanced credits. As will be seen, many forms of enhancement are possible.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0016] FIG. 1 is a system diagram illustrating various components of a gaming system according to embodiments of the system;

[0017] FIG. 2 is a functional block diagram that illustrates an example gaming machine that can be a part of the gaming system shown in FIG. 1;

[0018] FIG. 3 is a block diagram of an example bonus controller shown in FIG. 1 according to embodiments of the system;

[0019] FIGS. 4A and 4B together show a block diagram of the key components on the bridge shown in FIG. 1 for present embodiments of the system showing the main connections between each of the key components on the bridge;

[0020] FIGS. 5A and 5B together depict the microcontroller shown in FIG. 4B, which monitors SAS communications and provides Ethernet switch logic;

[0021] FIG. 6 shows the isolation circuit of FIG. 4B;

[0022] FIGS. 7A and 7B illustrate the game jacks (and related components) shown in FIG. 4B;

[0023] FIG. 8 depicts an Ethernet switch, not shown in FIGS. 4A and 4B, that is used to control signals on the game jacks of FIGS. 7A and 7B.

[0024] FIGS. 9A, 9B, and 9C together depict Ethernet switch controllers, not shown in FIGS. 4A and 4B, that are used to control signals on the Ethernet switches shown in FIGS. 8 and 13 for the game jacks (FIGS. 7A and 7B) and the network jacks (FIGS. 10A, 10B, and 11), respectively;

[0025] FIGS. 10A, 10B, and 11 depict the network jacks of FIG. 4B;

[0026] FIG. 12 depicts serial drivers, not shown in FIGS. 4A and 4B, for LED indicators on the network jacks of FIGS. 10A, 10B, and 11;

[0027] FIG. 13 depicts an Ethernet switch, not shown in FIGS. 4A and 4B, that is used to control signals on the network jacks shown in FIGS. 10A, 10B, and 11;

[0028] FIGS. 14 and 15 show the connectors for the single board computer shown in FIG. 4A;

[0029] FIG. 16 illustrates the tamper detection device shown in FIG. 4A;

[0030] FIGS. 17-20 show the components on the gaming machine interface board shown in FIG. 1 for present embodiments of the system;

[0031] FIG. 21 is a highly schematic depiction of the architecture of one embodiment of the present system;

[0032] FIG. 22A is a functional block diagram that illustrates a gaming device according to embodiments of the invention;

[0033] FIG. 22B is an isometric view of the gaming device illustrated in FIG. 22A;

[0034] FIG. 23A-23C are detail diagrams of exemplary types of gaming devices;

[0035] FIG. 24 is a functional block diagram of networked gaming devices;

[0036] FIG. 25 shows a block diagram of an example method;

[0037] FIG. 26 shows a block diagram of an example method; and

[0038] FIG. 27 shows a block diagram of an example system.

DETAILED DESCRIPTION

[0039] FIG. 1 is a system diagram illustrating various components of a gaming system. The gaming system 2 includes several electronic gaming machines (EGMs) 10, similar to EGM 200 in FIG. 2, that are each ultimately connected to a network 11. Each of the EGMs 10 of FIG. 1 connect to the network 11 (which may be a wireless network, a wired network, the Internet, an Ethernet network, combinations thereof, and the like) through first, a gaming machine interface (GMI) 12 and then a bridge control circuit (bridge) 14. As can be seen, each GMI 12 is further connected to a bridge 14, each bridge configured to accommodate one or more GMI connections, via their respective cables 13. In the present embodiments, up to 8 EGMs may be connected to each bridge. It should be appreciated that the number of connections is merely a matter of design, and more or less could be connected. For example, a single bridge may accommodate up to 64 EGMs, which would enable a single bridge to serve all of the machines in most all slot machines banks. It should be noted that not all of the jacks must be used. In other words, fewer than 8 GMIs (or whatever the maximum number is) can be connected to each bridge. The dots adjacent the groups of EGMs in FIG. 1 signify additional EGMs each with a corresponding GMI connected to a bridge, but to simplify the

drawing only 3 machines in each group are depicted. A bridge box may be contained in the cabinet of one of the EGMs, in a base that supports an EGM, or in another suitable location.

[0040] A conventional bonus controller **16** may be connected directly to some of the bridges **14** and may communicate over the network to any of the EGMs so connected in the network. Note that the bridges may communicate via an Ethernet protocol through the bonus controller or may bypass the bonus controller as shown for bridge **14** in the middle group of EGMs in FIG. **1**. In addition, bridges **14** can be connected in a daisy chain configuration, as shown via Ethernet link **17** in FIG. **1**. The links need not necessarily be Ethernet and the present disclosure contemplates wireless communications as well. The bonus controller **16** generally communicates through a non-SAS protocol, such as another well-known communication protocol known as Gaming Standards Association (GSA). GSA is typically carried over an Ethernet network, and thus the bonus controller **16** includes an Ethernet transceiver, which is described with reference to FIG. **3** below. Because the bonus controller **16** communication may be Ethernet based, a switch **18** may be used to extend the number of EGMs that may be coupled to the bonus controller **16**. The bonus controller **16** and/or the bridge **14** may create or convert data or information received according to a particular protocol, such as SAS, into data or information according to another protocol, such as GSA. In this way the bridge **14** and bonus controller **16** are equipped to communicate, seamlessly, between any EGM **10** and network **11** no matter which communication protocols are in use. Further, because the bridge **14** and bonus controller **16** are programmable, and include multiple extensible communication methods, as described below, they are capable of communicating with EGMs **10** that will communicate using protocols and communication methods developed in the future. The functions implemented by any of the controllers or processors mentioned herein might be distributed among a plurality of controllers or processors.

[0041] Other games or devices on which games may be played are connected to the gaming network using other connection and/or communication methods. For instance, a player kiosk **20** may be directly coupled to the gaming network. The player kiosk **20** allows players, managers, or other personnel to access data on the network **11**, such as a player tracking record, and/or to perform other functions using the network. For example, a player may be able to check the current holdings of the player account, transfer balances, redeem player points for credits, cash, or other merchandise or coupons, such as food or travel coupons, for instance.

[0042] A wireless transceiver **22** may couple the network **11** to a wireless EGM **24**, such as a handheld device, or, through a cell phone or other compatible data network, the transceiver **22** may connect to a cellular phone **26**. The cellular phone **26** may be a "smart phone," which in essence is a handheld computer (e.g., mobile computing device) capable of playing games or performing other functions

[0043] The network **11** also couples to the Internet **28**, which in turn is coupled to a number of computers, such as the personal computer **30**. The personal computer **30** may be used much like the kiosk **14**, described above, to manage player tracking or other data kept on the network **11**. More likely, though, is that the personal computer **30** is used to play actual games in communication with the network **11**. Player data related to games and other functions performed on the personal computer **30** may be tracked as if the player were playing on an EGM **10**.

[0044] In general, in operation, a player applies a starting credit into one of the EGMs, such as an EGM **10**. For example, the EGM **10** sends data through its SAS or other data communication port through the GMI **12** and associated bridge **14** to the network **11**. Various servers **32** and databases **34** collect historical accounting information about the gameplay on the EGMs **10**, such as wagers made jackpots won, gambling patterns, other historical gaming data, combinations thereof, and the like, for example. And, as will be described in more detail, the various servers **32** and databases **34** may continue to provide some services for the EGMs while the present embodiments take over others of the services. As will also be described in more detail, this feature permits any of the services offered by a legacy system, including bonus controller **16**, to be taken over by the present

system.

[0045] In addition, each EGM **10** may accept information from systems external to the EGM itself to cause the EGM **10** to perform other functions. For example, these external systems may cause the EGM **10** to issue additional credits to the player. In another example, a promotional server may cause the EGM **10** to print a promotional coupon on the ticket printer of the EGM.

[0046] The bonus controller **16** is configured to perform some of the above-described functions as well. For example, in addition to standard games on the EGM **10**, the bonus controller **16** is configured to drive the EGM **10** to pay bonus awards to the player based on any of the factors, or combination of factors, related to the EGM **10**, the player playing the EGM **10**, particular game outcomes of the game being played, or other factors.

[0047] In this manner, the combination of the bonus controller **16** and bridges **14** are a sub-system capable of interfacing with each of the EGMs on a network **11**. As a result, each bridge **14** may gather data about the game, gameplay, or player, or other data on the EGM **10**, and forward it to the bonus controller **16**. The bonus controller **16** may use such collected data as input and, when certain conditions are met, send information and/or data to the EGM **10** to cause it to perform certain functions.

[0048] In a more detailed example, suppose a player is playing an EGM **10** coupled to the bridge **14** and the bonus controller **16** described above. The player inserts a player tracking card so the EGM or any other devices connected to the network **11** may determine the player identity. The bridge **14** may be configured to store such identifying information. The bridge may pass such information to bonus controller **16**, which may be configured to provide a welcome-back bonus to any level-2 player after he or she has played two games. Gameplay on the EGM **10** may continue and, after the player plays two games, bonus controller **16** may cause EGM **10** to add an additional **40** credits to EGM **10** as the welcome-back bonus. Such monitoring and control of EGM **10** can occur in conjunction with, but completely separate from any player tracking or bonusing function that is already present on the network **11**. In other words, the bonus controller **16** may be configured to provide a time-based bonus of **10** credits for every hour played by the player of the EGM **10**. The above-described welcome-back bonus may be managed completely separately through the bonus controller **16** and bridge **14**. Further, all of the actions on the EGM **10** caused by the bonus controller **16** may be communicated to the standard accounting, tracking, and other systems already present on the network **11**. Additionally and/or alternatively, the welcome-back bonus described above may be implemented on servers **32** along with other functions.

[0049] With reference back to FIG. **1**, the bonus controller **16** may couple to each of the bridges **14**, and by extension to their coupled EGMs to cause data and commands to be sent to the EGMs to control functions on each EGM. FIG. **3** is a detailed block diagram of such a bonus controller.

[0050] Consideration will now be given to the structure and operation of bonus controller **16** before providing a general overview of the operation of system **2**. A bonus controller **300** of FIG. **3** may be an embodiment of the bonus controller **16** illustrated in FIG. **1**. The bonus controller **300** may comprise a microprocessor **310**. For example, the microprocessor **310** may be an Atmel AT91SAM9G20. The microprocessor **310** may be coupled to one or more memory systems **320**, **325**. As a non-limiting example, the memory system **320** may be a 2 Megabyte FRAM while memory system **325** may be a 64 Megabyte Synchronous DRAM (SDRAM). Of course, the sizes and types of memory included in any bonus controller may be determined to suit a particular implementation. For example, the memory may be computer readable media in the form of volatile memory, such as random access memory (RAM), and/or non-volatile memory, such as read only memory (ROM)

[0051] The microprocessor **310** may couple to one or more card readers, **340**, **345**, which may be configured to accept easily replaceable, portable memory cards. Each card reader of the one or more card readers may further include Electro-Static Discharge (ESD) devices configured to prevent damage to internal circuitry, such as the microprocessor **310**, when cards are inserted or

removed from the card readers **340**, **345**. In practice, a card in one of the card readers **340**, **345** may store program code for the microprocessor **310** while a card in the other reader may store data for use by the bonus controller **300**. Alternatively, a single card in either of the card readers **340**, **345** may store both program and data information.

[0052] A port connector **330** includes multiple communication ports for communicating with other devices. The communication processor of each bridge **14** may couple to a connected bonus controller through such a communication port. The communication port **330** may be, but is not necessarily an Ethernet interface, as described above, and may comprise a MAC address **331**. The port connector **330** may comprise multiple separate connectors, such as eight, each of which connect to a single bridge **14** (FIG. **1**), which in turn may connect to up to eight separate EGMs **10**. Thus, a single bonus controller **300** may couple to sixty-four separate EGMs by connecting through appropriately connected bridges.

[0053] Further, a second port connector **335** may be included in the bonus controller **300**. The second port connector may also be, but is not necessarily, an Ethernet connector. The second port connector **335** may be configured to allow additional connectivity to the bonus controller **300**. The second port connector **335** may couple to another bonus controller **300** or to other server devices, such as the server **60** on the network **11** of FIG. **1**. The second port connector **335** may additionally be coupled to a SMIB, thus providing the bonus controller **300** with the ability to directly connect to nine SMIBs.

[0054] Ethernet connections are easily replicated with a switch, external to the bonus controller **300** itself, which may be used to greatly expand the number of devices to which the bonus controller **300** may connect.

[0055] Systems to protect the integrity of the bonus controller **300** are included. For example, an intrusion detection circuit **360** may be configured to signal the processor **310** if a cabinet or housing that contains the bonus controller **300** is breached, even if no power is supplied to the bonus controller **300**. The intrusion detection circuit may comprise a magnetic switch that closes (or opens) when a breach occurs. The microprocessor **310** may generate a signal that may be detected by one or more devices connected to the network **11**. The signal may indicate that a breach occurred and may cause an appropriate response. An on-board power circuit **370** may provide power to the bonus controller **300** for a relatively long time, such as a day or more, so that any data generated by the processor **310** is preserved and so that the processor **310** may continue to function, even when no external power is supplied. The on-board power circuit **370** may comprise an energy-storing material such as a battery or a capacitor.

[0056] Similarly to the microprocessor processor **260** of the SAS processor **210** described above, the microprocessor **310** of the bonus controller **300** may be coupled to a program/debug port for programming the microprocessor **310**. For example, program and/or other data for the microprocessor may be updated through the program/debug port.

[0057] In operation the bonus controller **300** may configure and control bonus features on gaming machines through network **11** or through other communication systems. Bonus features may be implemented through each gaming machine's internal structure and capabilities, and may include integration with additional peripheral devices (e.g., mobile computing devices such as smart phones). Bonusing programs for the connected games may be introduced to the bonus controller **300** by updating data stored in the memory systems directly on the bonus controller, or by inserting new memory cards in one or more of the card readers **340**, **345**. Such a platform provides a way for game developers, even third-party developers, to define and program new types of bonus games that may be used in conjunction with existing EGMs on existing gaming networks, or on new games and new networks as they are developed.

[0058] Before providing an overview of embodiments of the system disclosed herein, consideration will first be given to an electronic gaming machine.

[0059] FIG. **2** is a functional block diagram that illustrates an example electronic gaming machine

(EGM) **200**. The system may comprise one or more EGMs. The one or more EGMs may comprise one or more types of electronic gaming machines, such as physical reel slot machines, video slot machines, video poker gaming machines, video blackjack machines, keno games, and any other type of devices may be used to wager monetary-based credits on a game of chance.

[0060] The illustrated gaming machine **200** may comprise a cabinet **205** configured to house various parts of the gaming machine **200**, thereby allowing certain components to remain securely isolated from player interference, while providing access to player input/output devices so that the player may interact with the gaming machine **200**. The securely housed components may comprise a game processor **210**, memory **215**, and connection port **250**. The game processor **210**, depending on the type of gaming machine **200**, may completely or partially control the operation of the gaming machine. For example, if the gaming machine **200** is a standalone gaming machine, game processor **210** may control virtually all the operations of the gaming machine and attached equipment. In other configurations, the game processor **210** may implement instructions generated by or communicated from a remote server or another controller. For example, the game processor **210** may be responsible for running a base game of the gaming machine **200** and executing instructions received over the network from a bonus server or player tracking server. In a server-based gaming environment, the game processor **210** may simply act as a terminal to perform instructions from a remote server that is running game play on the gaming machine **200**. The functions implemented by the processor might also be distributed among several processors.

[0061] The memory **215** is connected to the game processor **210** and may be configured to store game information about gameplay or player interactions with the gaming machine **200**. This memory may be volatile (e.g., RAM), non-volatile (e.g., flash memory), or include both types of memory. The connection port **250** is also connected to the game processor **210**. This connection port **250** typically connects the gaming machine **200** to a gaming network (e.g., the network **211**). The connection port **250** may be configured as a serial port, parallel port, Ethernet port, optical connection, wireless antenna, or any other type of communication port used to transmit and receive data. Although only one connection port **250** is shown in FIG. 2, the gaming machine **200** may include multiple connection ports. Virtually all gaming machines, however, permit transfers of cash or promotional credits between the slot accounting system and the machine via a single designated port. As described above, in many existing gaming machines, this connection port **250** is a serial connection port utilizing a SAS protocol to communicate with one or more remote game servers, such as player tracking servers, bonus servers, accounting servers, etc.

[0062] The player input/output devices housed by the gaming cabinet **205** further include a bill/ticket reader **270**, a credit meter **285**, and one or more game speakers **295**. Various gaming machines may include fewer or more input/output devices (e.g., a game handle, a coin acceptor, a coin hopper, etc.) depending upon the configuration of the gaming machine.

[0063] The gaming machine **200** may comprise a gaming display **220**. The gaming display **220** may comprise one or more mechanical spinning reels, a video display, combinations thereof, and the like. The gaming display may be configured to display aspects of the gameplay to the player. The gaming display **220** may comprise a touchscreen configured to receive one or more user inputs. For example, if the gaming display **220** is a video display, the gaming display may include a touch screen to further allow the player to interact with game indicia, soft buttons, or other displayed objects. The gaming machine **200** may comprise a button panel **230**. The button panel **230** may be configured to allow the player to select and place wagers on the game of chance, as well as allow the player to control other aspects of gaming. For example, some gaming machines allow the player to press one of buttons **233** to signal that he or she requires player assistance. Other buttons may bring up a help menu and/or game information. The buttons **233** may also be used to play bonuses or make selections during bonus rounds.

[0064] The gaming machine may comprise one or more ticket printers **275**. The one or more ticket printers **275** may be configured to print one or more tickets. For example, as discussed in greater

detail herein, the one or more ticket printers **275** may be configured to print one or more tip tickets and/or one or more enhanced value tickets. The tickets can typically be redeemed for cash at a cashier cage or kiosk. The one or more ticket printers **275** may be configured to connect to the game processor and/or to a remote server, such as a TITO server to accomplish its intended purpose. In gaming machines that have more than one peripheral device, and which include only a single SAS port, the peripheral devices all share communication time over the connection port **250**. [0065] The gaming machine **200** may comprise a player club interface device **260**. The player club interface device **260** may include a reader device and one or more input mechanisms. The card reader device is configured to read an object or indicia identifying the player. The identifying object may be a player club card issued by the casino to a player that includes player information encoded on the card. Once the player is identified by a gaming machine, the player club interface device **260** communicates with a remote player server (e.g., player club server) through the connection port **250** to associate a player account with the gaming machine **200**. This allows various information regarding the player to be communicated between the gaming machine **200** and the player server, such as amounts wagered, credits won, rate of play, gambling habits, combinations thereof, and the like. In other embodiments, the card reader may read other identifying cards (such as driver licenses, credit cards, etc.) to identify a player. Although FIG. 2 shows the reader as a card reader, other embodiments may include a reader having a biometric scanner, PIN code acceptor, QR code scanner, or other methods of identifying a player to pair the player with their player tracking account.

[0066] Other input/output devices of the gaming machine **200** include may a credit meter **285**, a bill/ticket acceptor **270**, and speakers **295**. The credit meter **285** may be configured to indicate a total number of credits remaining on the gaming machine **200** that are eligible to be wagered. The credit meter **285** may be configured to indicate a number of credits eligible to transfer to another player or transfer to an attendant as a tip. The credit meter **285** may reflect a monetary unit, such as dollars, or an amount of credits. The bill/ticket acceptor **270** may be configured to recognize and/or validate currency (e.g., paper bills) and/or printed tickets and may be configured to cause the game processor **210** to display a corresponding amount on the credit meter **285**. The speakers **295** may be configured to play auditory signals in response to game play and/or may be configured to play enticing sounds while in an “attract-mode,” when a player is not at the gaming machine. The auditory signals may also convey information about the game, such as by playing a particularly festive sound when a large award is won. The auditory signals and/or any visual outputs may be associated with one or more paytables and or one or more payback variances. For example, the one or more auditory signals or visual outputs may be varied (e.g., adjusted) based on a payback variance associated with the gaming machine. For example, the attract mode may be enhanced when the machine has not been played and a variance with a predetermined payback percentage exists.

[0067] The player may initially insert monetary bills or previously printed tickets with a credit value into the bill acceptor **270**. The player may also put coins into a coin acceptor (not shown) or a credit, debit, or casino account card into a card reader/authorizer (not shown). In other embodiments, stored player points or special ‘bonus points’ awarded to the player or accumulated and/or stored in a player account may be able to be substituted at or transferred to the gaming machine **200** for credits or other value. For example, a player may convert stored loyalty points to credits or transfer funds from his bank account, credit card, casino account or other source of funding. The selected source of funding may be selected by the player at time of transfer, determined by the casino at the time of transfer or occur automatically according to a predefined selection process.

[0068] The gaming machine **200** may comprise various other devices to interact with players, such as light configurations, top box display **290**, and secondary display **280**. The top box display **290** may include illuminated artwork to announce a game style, a video display (such as an LCD), a

mechanical and/or electrical bonus display (such as a wheel), or other known top box devices. The secondary display **280** may be a vacuum fluorescent display (VFD), a liquid crystal display (LCD), a cathode ray tube (CRT), a plasma screen, or the like. The secondary display **280** may be configured to display any combination of primary game information and ancillary information. For example, the secondary display **280** may show player tracking information, secondary bonus information, advertisements, or player selectable game options. The secondary display may be attached to the game cabinet **205** or may be located near the gaming machine **200**. The secondary display **280** may also be a display that is associated with multiple gaming machines **200**, such as a bank-wide bonus meter, or a common display for linked gaming machines.

[0069] In operation, typical play on a gaming machine **200** commences with a player placing a wager on a game to generate a game outcome. The wager may comprise one or more credits and/or currency. In some games, a player need not interact with the game after placing the wager and initiating the game, while in other games, the player may be prompted to interact with the gaming machine **200** during game play. Interaction between the player and the gaming machine **200** is more common during bonuses, but may occur as part of the game, such as with video poker. Play may continue on the gaming machine **200** until a player decides to cash out or until insufficient credits remain on the credit meter **285** to place a minimum wager for the gaming machine.

[0070] Before going into detail about the structure and operation of bridge **14** and gaming machine interface **12**—as well as a detailed description of the overall operation of system **2**—consideration will first be given to a high-level view of the operation of system **2**. As mentioned above, prior art systems for providing services to and management functions for networked gaming machines suffer from a number of problems. One of the ways system **2** addresses these problems is by providing the services, such as TITO, dispatch, bonuses, etc. from the bridge circuit as opposed to a server in a secure IT room.

[0071] TITO services are described in U.S. Pat. Nos. 5,265,874; 5,290,033; 6,048,269; and 6,729,957, all of which are incorporated herein by reference. Dispatch services are described in US Patent Application Publication 2017/0186270, which is also incorporated herein by reference.

[0072] As will be discussed in detail, each bridge circuit has a SAS processor and a network processor. The SAS processor handles SAS communications, which is a commonly used protocol for communicating via a gaming machine port. The network processor handles communication between and among the bridges, the bonus controller—and any other devices connected to network **11**—and ultimately to the servers **32** and databases **34**. When all services, such as TITO, dispatch, bonuses, etc., are implemented by processes on bridge **14**, servers **32** may function only to collect and store (in databases **34**) historical data sent from flash memory in each bridge. This is in contrast to prior art systems in which the processes that provide various services to the EGMs are located on the servers in a secure room.

[0073] The SAS processor in bridge **14** may be configured to communicate with a SAS port on the EGM to which it is connected via GMI **12**. The GMI **12** may be configured to convert signals from the SAS port of the EGM to the commonly used RS232 standard for two-way serial communication over a USB connection to bridge **14**. The bridges, bonus controller, switches, (e.g., switch **18** in FIG. **1**), and all the other devices, including servers **32**, may communicate with one another via the Ethernet protocol or a wireless protocol. The bridge includes a network processor that may be configured to communicate with both the SAS processor in the bridge, via USB, and with network **11**, via Ethernet, to relay information to and from the SAS processor.

[0074] In operation, bridge **14** conducts high-speed polling to retrieve data that appears at the EGM SAS port. Unlike prior art systems, which typically poll only for EGM meter readings, bridge **14** requests and receives extensive information related to any activity on the EGM. For example, each actuation of a button on the EGM and each reel stop during play of a game are collected, each with its own time stamp. The collected data may be stored in a memory in bridge **14**, and may also be sent to flash memory in each of two additional bridges, which functions as a backup system. One

manner in which the collected data can be used is described in applicant's U.S. Pat. No. 10,553,072 issued on Feb. 4, 2020, invented by John Acres, which patent is incorporated herein by reference for all purposes.

[0075] Computer programs that provide the various services, such as TITO cashless tickets, progressive jackpots, player tracking, and other services may be executed on the SAS processor or on network processor **42**. Although all required services may be run by the bridge, if a legacy system that includes a suite of programs to provide services remains in place, selected services can be run on the bridge while letting other of the services continue to be run by the legacy system. This permits the present system to take over services provided by one or more other vendor systems one at a time, if that is desired. Ultimately, the bridge can take over all management and EGM services.

[0076] The bridge **14** may comprise a robust power management system. For example, the bridge **14** may comprise one or more large capacitors that provide several functions. For example, in the event of a brownout, a power failure that lasts less than about 30 seconds, all power to bridge **14** and GMI **12** may be supplied from the capacitors. As a result, no functionality is lost during the brownout. For example, in the event of a longer outage, data for transactions in process may be stored, and when power is resumed the transaction may be completed. For example, the SAS processor can run for as many as three days in the event power is lost. This enables the processor to continue to monitor **14** tampering during a prolonged power outage. For example, if cable **13**, which connects bridge **14** with one of its associated GMIs **12**, is unplugged, GMI **12** may be supplied with power via a capacitor configured to permit an orderly shutdown of GMI **12** to preserve data in a current transaction.

[0077] The software programs in the bridge may have high-speed access to data as soon as it is stored in the flash memory. The software programs may be provided by the owner of the hardware system described herein or by third parties.

[0078] Data may be rapidly moved upstream from the flash memory to a repository managed by a server where specialized software captures, indexes, and correlates real-time, machine-generated data. Software sockets may be used to monitor specific data in the repository. As a result, the repository can be used by third parties, each of whom are granted secure access, to implement any of the services that might be required on the casino floor, including bonus games, progressive jackpots, TITO services, accounting and reports, etc. For example, the present system can provide data in the repository within a typical range of about 50-150 milliseconds from the time the event occurs on the gaming machine. This contrasts with prior art systems that provide data, but not all of the data, to a database where the lag time from the event on the machine to appearance in the database is typically in the range of 1-3 seconds. This delay prevents or impairs implementing robust services using the prior art systems.

[0079] Security is maintained via a novel system of manufacturing and venue certificates and tokens as well as a tamper-proof feature for each bridge box.

[0080] Consideration will now be given to the bridge hardware devices and how they are connected to one another. In FIGS. **4A** and **4B**, a block diagram illustrates the various components and functionalities provided by the bridge **14**. Additional detail will be provided in later description and drawings for some of these, which are shown only generally in the FIG. **4** overview.

[0081] Bridge circuit **14** in FIG. **4B** may comprise one or more game jacks (e.g., 8 game jacks), indicated generally at **36** in FIG. **4B**. Each of the game jacks may comprise a receptacle for an RJ45 jack. Cable **13** (in FIG. **1**), which is not shown in FIGS. **4A** and **4B**, connects each GMI **12** to its associated bridge **14** via RJ45 plugs on either end of the cable. Each GMI **12** is in turn connected to a SAS port on its associated EGM as will be later described in more detail. Each game jacks of the one or more game jacks **36** may comprise one or more pins configured to accommodate RS245 serial communications as well as pins to accommodate 100 MB Ethernet communications between GMI **13** and bridge **14**

[0082] Considering the one or more game jacks **36** in more detail, attention is directed to FIGS. **7A** and **7B**. As shown in the figures, each game jack of the one or more game jacks **36** is connected to an Ethernet transformer **44**, each transformer handling communications to and from two EGMs. The Ethernet 100 MB signal lines for each EGM are shown generally at **46** and are available for expansion of services to the EGM. Electrostatic discharge protection for Ethernet signal lines **46** is provided by components **48**. Each game jack of the one or more game jacks **36** may comprise one or more signal lines indicated generally at **50** configured to accommodate the RS245 signals from the GMI that the one or more game jacks are connected to. The one or more signal lines may be configured to carry SAS data from the EGM. Each EGM has 2 signal lines, (e.g., EGM36 in FIG. **7B** has lines labeled EGM8.5 and EGM8.4 to indicate the third of eight potential EGM connections to bridge **14**), with the connections being on pins **4** and **5** of each jack. Each of the other 7 EGM connections are similarly labeled. Each of the EGM RS245 signal lines is connected to a galvanic isolation circuit **43** in FIG. **4B** and each line bears the same label in FIG. **4B** as in FIGS. **7A** and **7B**.

[0083] The one or more network jacks **38**, are used to connect the bridge to network **11**. The one or more network jacks may be configured to connect the bridge **14** to the network **11** via one or more Ethernet connection. In addition, there is a jack **39** in FIG. **4B** that is one end of a connection that is internal to bridge **14**, which will be shortly described. The one or more network jacks **38** may be used in a variety of ways. For example, two of the network jacks may be connected to the network to provide redundancy in the event that communication over one of the connections (e.g., a network jack of the two network jacks) is impaired. One network jack may be configured to connect to a peripheral device, such as bonus controller **16** in FIG. **1**. In addition, bridges **14** can be daisy chained together to facilitate communications with network **11**, also shown in FIG. **1**.

[0084] Considering the one or more network jacks **38** in more detail, in FIGS. **10A** and **10B**, show all **6** jacks labeled A through F. Each of the jack lines internal to bridge **14** are connected to electrostatic protection devices **39**. Each of the lines are labeled as shown at **45**. These lines connect to a correspondingly labeled pins on an Ethernet switch in FIG. **13**.

[0085] In addition, jack **39** (also shown in FIG. **4B**), which is shown in more detail in FIG. **11**, has internal lines labeled as shown that are connected to correspondingly labeled lines on the Ethernet switch of FIG. **13**. Jack **39** is connected to one end of a cable **40**, in FIGS. **4A** and **4B**. The other end of the cable is plugged into a jack that is provided on network processor **42** board.

[0086] As a result, this cable connects the network processor to the network Ethernet switch, depicted schematically in FIGS. **4A** and **4B**.

[0087] In FIG. **12**, shift registers **45** are connected to LEDs on each of network jacks **38** as shown. The shift registers also connect to lines LED.CK and LED.DA, which in turn are connected to the correspondingly labeled lines on an Ethernet switch controller **56** in FIG. **9C**. These signals on these lines ultimately light LEDs on each network jack to indicate connection status and when data is moving over the connection.

[0088] In the present embodiments, network processor **42** may be a credit-card size, 64-bit, single-board computer with both USB and Ethernet ports. The current embodiment may use the ROCK64 single-board computer manufactured by PINE64, but any suitable single-board computer could be used. As a result of these connections, data in the bridge is transmitted to and from the Ethernet switch in FIG. **13** and from there to and from network **11**.

[0089] Returning again to FIG. **4B**, the SAS data (shown in more detail at **50** in FIGS. **7A** and **7B**) on the RS485A and RS485B lines from each of game jacks **36** are connected to one side of a galvanic isolator circuit **43** shown in FIG. **4B** and in more detail in FIG. **6**. In the present embodiment, isolation is provided for each signal pair from the game jacks **36**, like signal pair RS485.5A and RS485.5B in FIG. **6**, by galvanically-isolated transceivers **52**. In the present embodiments, transceivers **52** may comprise model number ISO1410 manufactured by Texas Instruments. The other side of each transceiver **52** generates a TTL signal on 4 lines as shown and

is connected to a microcontroller **53**, shown generally in FIG. **4B** and with more detail in FIGS. **5A** and **5B**.

[0090] In the present embodiments, microcontroller **53** comprises a SAM E53 microcontroller manufactured by Microchip. Each of the four lines of transceivers **52** is connected to microcontroller **53** via the correspondingly labeled lines in FIGS. **5A** and **5B**, on the one hand, and in FIG. **6**. These connections are also shown in the diagram of FIG. **4B**.

[0091] Microcontroller signal lines are also connected to Ethernet switch controllers **54**, **56** in FIGS. **9A** and **9C**. In the present embodiments, each switch controller is a Realtek RTL8370MB-CG switch controller. Switch controller **54** controls 100 MB Ethernet communications, and switch controller **56** controls 1G Ethernet communications.

[0092] Both switch controllers **54**, **56** are connected to microcontroller **53** in FIGS. **5A** and **5B** via the labeled lines shown at the bottom of FIG. **9B**. The labels at the bottom of FIG. **9B** correspond to labels on microcontroller **53** lines shown in FIGS. **5B**. These connections are also shown in FIG. **4B** at the lower portion of the microcontroller marked Ethernet Switch Logic. In addition, Ethernet switch controller **56** is connected to microcontroller **53** as shown in FIG. **9C** in the block labeled RMII Interface (Reduced Media-Independent Interface), although these connections are not shown in FIGS. **4A**, **5A**, and **5B**.

[0093] Turning again to FIGS. **4A** and **4B**, microcontroller **53** includes a line **58** that goes to one of the voltage regulators **60** in FIG. **4A**. These include a 5V regulated power supply (not shown in the drawings). The 5V supply supplies network processor **42**, which requires 5 volts at 3 amps. A signal on line **58** can be used to shut this power supply down under conditions that will be further explained. There is also a regulated 3V power supply for microcontroller **53**, which draws considerably less current than network processor **42**. In addition, there is an isolated regulated 5V power supply drawing 1 amp that supplies power to each GMI **12** and a 3.3V at 1-amp supply and 1V at 1-amp supply for the Ethernet switches.

[0094] Power may be provided by power supply **61** in FIG. **4A**, which is a 40-Watt supply that provides 12 volts to the voltage regulators. In the present embodiments the power supply is manufactured by XP Power under model number VEC40US12. The power supply plugs into a conventional power outlet.

[0095] Beneath voltage regulators **60** in FIG. **4A** is a tamper detection circuit **62**, which is shown in more detail in FIG. **16**. In the present embodiments, the circuit comprises a switch **63** that includes a magnetic ferrous ball **64**, which adheres to a plate **66** that is part of the container for ball **64**. The plate is relatively flush against a lid **68** of the metal box (not otherwise shown) that contains bridge **14**. When the lid is lifted to gain access to bridge **14**, the ball drops onto a contact **70**, which completes a circuit between the container for ball **64**, which is grounded as shown, and contact **70**, which is connected to the TAMPER line in FIG. **4A**. This provides a low-state signal to the microcontroller, which in turn generates and sends a signal configured to destroy a cryptographic key in microcontroller **53**. The key may be configured to use in encrypting and decrypting data sent to and from bridge **14**. Destruction of the key under these circumstances prevents further operation of bridge **14** and the associated GMIs **12** until the box containing the bridge can be inspected (either by the manufacturer or the appropriate gaming authority) to confirm whether or not tampering occurred. If tampering occurred, a new encrypted key may be delivered over the network to return bridge **14** and its associated GMIs **12** to service.

[0096] Beneath tamper detection circuit **62** is a temperature monitoring circuit **72**. And beneath temperature monitoring circuit **72** is a super-capacitance (supercap) control circuit **74**, which will be described in more detail after description of bridge **14** and GMI **12**.

[0097] Before considering operation of the supercap control circuit **74**, consideration will be given to the manner in which network processor **42** is connected to other components in bridge **14**. As previously mentioned, plugs on either end of cable **40** connect at one end to the Ethernet switch in FIG. **13** and at the other end to network processor **42**. This connection is shown diagrammatically

in FIGS. 4A and 4B. This connects network processor 42 to network 11. Network processor 42 is also connected to microcontroller 53 via a cable that includes USB, PIO (Programmed Input/Output), and 5V power, as shown in FIG. 4A. The jack on bridge 14 that enables this connection is shown at 76 in FIG. 14 with pin labels that correspond to those in FIG. 4A. The FIG. 4A diagram shows these connections in box 75 marked SBC Connect (SBC is single board computer, namely network processor 42). As can be seen on jack 76 in FIG. 14, each of the labeled lines is connected to a correspondingly labeled line in box 75, except for PFO, which is tied to the 5V line in FIG. 14. Two of the connections, namely MON.USB.DM on line 78 and MON.USB.DP on line 80 in FIG. 14 are filtered via a single pair common mode filter 82. The signals on lines 78, 80 are shown with corresponding labels in box 75 in FIG. 4A, which provide USB communication between network processor 42 and microcontroller 53.

[0098] FIG. 15 is a second jack having similar connections to jack 76 in FIG. 14. The jack in FIG. 15 may be used for diagnostics.

[0099] Consideration will now be given to the structure of GMI 12 and how it is connected to its associated EGM and to its associated bridge 14. GMI 12 is shown in several drawings, FIGS. 17-20. GMI 12 may be formed on two boards, one of which may be a motherboard, indicated generally at 77 in FIG. 20. The motherboard includes 3 jacks 78, 79, 80. Jack 78 may be configured to receive a plug (not shown) on one end of a cable 13, in FIG. 1. The other end of cable 13 may include a plug that is connected to any one of game jacks 36 in FIG. 7A and 7B, thus connecting GMI 12 to bridge 14.

[0100] Jack 79 may comprise an expansion port. Jack 79 may be available to connect to equipment on an EGM, such as a player tracking unit, a bonus feature, or any other auxiliary device that could be implemented using communications over network 11. The pins in jack 79 may be tied to the pins in jack 78 and therefore may communicate with the pins in the associated gaming jack 36 to which cable 13 is connected in the same manner as jack 78.

[0101] Jack 80 may be configured to provide communication between motherboard 77 and a daughterboard, the components of which are shown in FIGS. 17-18. One end of a cable (not shown) is plugged into jack 80 and the other end of the cable is plugged into a jack 82, in FIG. 19, on the daughterboard. The signal labels shown on the pins of jack 82 are connected to correspondingly labeled lines in FIG. 17.

[0102] A brief description of a typical gaming network into which the present embodiments are installed will aid understanding of how GMI 12 is further connected. As mentioned above, prior art systems for managing and providing services to EGMs typically include a suite of programs that deal with such things as accounting, cashless tickets, loyalty, and audit functions. The prior art system is referred to herein as a legacy host system. Communication with each EGM and the network that provides these services is made through the primary SAS port of each EGM. In prior art systems the primary SAS port is the only port that permits transfers of credits to and from the machine. Thus, in prior art systems, the SAS port must be used whenever money is transferred to and from the EGM, (e.g., bonus credits, TITO transactions, etc). As a result, the port to which GMI 12 must be connected is already connected to the network.

[0103] The current embodiments address this situation by connecting to network 11 in a manner that interposes bridge 14 between the EGM and the legacy host system. When installing GMI 12 where a legacy system is present, the cable from the legacy system that is plugged into the primary SAS port of EGM 10 is first unplugged. Interposing bridge 14 between the host system and EGM 10 may be accomplished via a jack 81 in FIG. 18, which receives one end of a connector (not shown) that plugs into jack 81. As a result, GMI 12 communicates with both the EGM, the signals of which are labeled EGM.TX (transmit) and EGM.RX (receive) in FIG. 18, and the host legacy system, the signals of which are labeled LEGACY.TX and LEGACY.RX in FIG. 18. These EGM and legacy signals are on lines on a daughterboard that are correspondingly labeled in FIG. 17.

[0104] Turning now to FIG. 17, a microcontroller 82 may be responsible for, among other things,

SAS communications with associated EGMs. In the present embodiments, microcontroller **82** comprises a SAMD51G18A microcontroller manufactured by Microchip Technology. The microcontroller includes internal memory, and it is also connected to a ferroelectric random-access memory **83**. Memory **83** may be an MR45V064B memory manufactured by Lapis Semiconductor Co. in the present embodiments. Memory **83** may be configured to retain data for up to 10 years without a source of power.

[0105] A line driver and receiver device, indicated generally at **84**, is connected to receive and transmit ports on microcontroller **82** in FIG. **17** as shown. Device **84** includes two receive drivers **84a**, **84c** and two transmit drivers **84b**, **84d**. These drivers are connected to transmit and receive lines for the EGM SAS port and the legacy system port via jack **81** in FIG. **18**. For example, device **84** may be a TRS3232 manufactured by Texas Instruments.

[0106] Microcontroller **82** may be configured as a SAS replicator. As a result, when the legacy system sends a SAS command, it is received at GMI **12** by microcontroller **82** via line driver **84c** in FIG. **17**. For example, if a player at the EGM presses the cashout button to receive a TITO ticket for the balance on the EGM credit meter, that command is received by line driver **84a** and passed to the legacy system via driver **84d**. When the legacy system determines all is in order, the legacy system assigns a ticket number for the TITO ticket and sends that data with a command to issue the ticket, which is received by line driver **84c**. When this happens, the SAS replicator acknowledges the command, and immediately confirms to the legacy system that it has been executed. The confirmation is sent by microcontroller **82** via the transmit line driven by line driver **84d**. Next, microcontroller **82** generates a SAS command to issue the ticket, which is sent to the EGM via line driver **84b**.

[0107] In the event of a power failure that happens between the time bridge **14** confirms issuance of the ticket to the legacy system and the time the command to issue the ticket is sent to the EGM via line driver **84B**, the data for this transaction is stored in memory (FeRAM) **83**. When power is restored, the data is retrieved and the transaction is completed. Further attention will be given to the manner in which bridge **14** responds to a power failure.

[0108] In this manner bridge **14** is interposed between the legacy system and the EGM. If the operator of the EGMs wishes to implement a TITO system provided by a vendor different from the vendor that provided the legacy system, the new TITO system can be implemented as a computer process run by network processor **42**. When that happens the legacy system may still remain in place because it may be providing other services, such as accounting and collecting player-tracking data. But legacy TITO commands received at driver **84c** are ignored because all TITO communications and commands are being transmitted via line drivers **84a**, **84b**.

[0109] Many of the new services, such as TITO and bonuses, can be provided via software installed at bridge **14**. This vastly reduces the regulatory burden. While the software must be approved, in many cases no additional hardware is required. This enables small design firms and even individuals to provide various products that require only new software.

[0110] In this way, various services, such as bonusing, dispatch, player-tracking, etc., can be implemented, one at a time, on bridge **14** while leaving the legacy system in place. An operator of EGMs may be interested in receiving EGM data to use for analytics and reporting. As previously mentioned, most prior art systems poll only for events, such as a cashout, hand-pay jackpot, etc., and current meter values. The present methods and systems poll rapidly and continuously to receive data that represents virtually all activity on the EGM, including reel stops, time between game play, and the like. If new data is detected, the present system generates a request to receive all available data. In other words, if the present system detects any available activity, it requests and polls for all available data. This provides a robust data stream for reporting and analytics.

[0111] Another line driver **85** includes a driver and receiver with electrostatic discharge protection. For example, the line driver **85** may comprise an SN65HVD72 half-duplex line driver and receiver manufactured by Texas Instruments. Line driver **85** may be connected to the R485.A and RS485.B

lines from jack **82** in FIG. **19**, which (as shown in FIG. **20**) may be connected via jack **80** to lines on jack **78**. As will be recalled jack **78** is connected to one of game jacks **36**. These are the transmit and receive lines between GMI **12** and bridge **14**.

[0112] The signal lines SWCLK and SWDIO may be lines configured to program flash memory contained in microcontroller **82**. These lines may be connected to the corresponding lines in jack **82** and from there via a cable (not shown) that connects the motherboard and daughterboard and ultimately via jack **78** to one of game jacks **36** on bridge **14**. This enables the flash memory in microcontroller **82** to be programmed with code delivered to it over network **11**.

[0113] The four lines that connect to memory **83** may be used to store transactions in process that are contained in a memory of microcontroller **82** in the event of a power failure, as will be more fully described.

[0114] The LED lines on the lower left of microcontroller **82** may be configured to drive LEDs that are diagnostic indicators to indicate the status of the communications between the EGM and GMI **12** and between the legacy host and GMI **12**.

[0115] The remaining components in FIG. **17** show power supply connections, biasing resistors, and capacitors, which are routinely used in connection with microcontrollers, like microcontroller **82**.

[0116] The present embodiments may incorporate a number of existing software programs to build the infrastructure of applications. For example, Consul and Nomad, both open-source programs provided by HashiCorp, may provide dynamic application coordination. For example, Vault, also open source and provided by HashiCorp, may provide the backbone for security and trust. For example, Kafka, an open source program provided by Apache Software Foundation, may provide asynchronous messaging and publish/subscribe functionality for communications from one bridge **14** to another. These programs may enable a single identified bridge **14** to be a master, e.g., in connection with issuing TITO tickets, and all the others to be slaves. For example, the master tracks and issues ticket numbers at the single bridge **14** and authorizes other bridges accordingly when a TITO ticket is authorized to be issued. If the master goes out of service, another board may be automatically selected to become the new master, thus providing seamless operation.

[0117] Consideration will now be given to various backup power supplies, which are each provided by large capacitors that are kept charged during normal operation, (e.g., when power supply **61** in FIG. **4A** is connected to power). There may be, for example, three sources of backup power that each provide a different function.

[0118] The first backup power supply is supercap control **74** in FIG. **4A**. Although not depicted in a drawing beyond FIG. **4A**, circuit **72** includes a high current supercapacitor backup controller and system monitor. For example, one manufactured by Linear Technology and identified as LTC3350EUHF. This device may be configured to control charging of 4 10-Farrad capacitors connected in series. Along with several power MOSFETs and biasing components, the Linear Technology device maintains a charge on the 4 capacitors. These capacitors back up the 5V and 3.3V regulated power supplies for microcontroller **53** and various other components.

[0119] These capacitors have enough stored charge to provide power to all components on bridge **14** for at least 30 seconds. If power is out for less than 30 seconds, or if power drops below normal levels for a short time, supercap control **74** may maintain power to all components on bridge **14**. As a result, normal operation continues without interruption for short power failures and brownouts.

[0120] If, however, the power failure extends beyond **30** seconds, supercap control **74** generates a signal that goes to microcontroller **53**. In response, microcontroller **53** launches a process that notifies network processor **42** that power is about to be lost. In response, bridge **14** and the GMIs **12** associated with the bridge begin a partial-shutdown operation and enter a hibernation mode during which a second backup power supply, indicated generally at **86** in FIG. **5B**, provides power to microcontroller **53** only. Power to network processor **42**, which draws a large current compared to microcontroller **53** is removed. In addition, any data in microcontroller **82** in FIG. **17** on the

GMI daughterboard is stored in memory (e.g., FeRAM) **83**.

[0121] Second backup power supply **86** may comprise a 1 amp diode **87** with a very low voltage drop. In the present embodiments, diode **87** may be a MAX40200 manufactured by Maxim Integrated. Second backup power supply **86** may also include a 45 Farad capacitor, which has enough charge to run microcontroller **53** for at least 100 hours. Its only function during this time is to detect a tamper signal from circuit **62**, in FIGS. **4A** and **16**. As will be recalled, the cryptographic key expires every 48 hours and a new encrypted key is delivered over the network. If the power is out long enough for the key to expire, no new key will be delivered. Or if the tamper circuit detects that the box containing bridge **14** is opened, the key will be destroyed.

[0122] In either of these circumstances, the box containing bridge **14** must be inspected, by either the manufacturer or an appropriate gaming authority, and reset to receive a new key when the box is reinstalled. This procedure provides a highly secure environment for data transactions handled by bridge **14**.

[0123] A third backup power supply may comprise a capacitor **88** in FIG. **20**. This capacitor maintains power to microcontroller **83** for a short time in the event cable **13** is unplugged. As will be recalled, cable **13** provides power to GMI **12**. When the voltage on capacitor **88** drops below a predefined value, microcontroller **82** shuts down in the same manner as when it receives a signal from bridge **14** indicating that power supply **61** is no longer providing sufficient power to run bridge **14**. As a result, if cable **13** is unplugged, any SAS data in microcontroller **82** is stored in memory (e.g., FeRAM) **83**. When power is restored, i.e., when cable **13** is plugged back in, that data is retrieved and microcontroller **82** resumes processing where it left off.

[0124] FIGS. **22A** and **22B** illustrate example gaming devices (e.g., gaming machines). Referring to FIGS. **22A** and **22B**, a gaming device **2210** is an electronic gaming machine. Although an electronic gaming machine or “slot” machine is illustrated, various other types of devices may be used to wager credits or other value on a game of chance. The term “electronic gaming device” is meant to include various devices such as electro-mechanical spinning-reel type slot machines, video slot machines, and video poker machines, for instance. Other gaming devices may include computer-based gaming machines, wireless gaming devices, multi-player gaming stations, modified personal electronic gaming devices (such as cell phones), personal computers, server-based gaming terminals, and other similar devices. For ease of illustration the present embodiments will be described in reference to the electronic gaming machine **2210** shown in FIGS. **22A** and **22B**.

[0125] The gaming device **2210** includes a cabinet **2215** housing components to operate the gaming device **2210**. The cabinet **2215** may include a gaming display **2220**, a base portion **2213**, a top box **2218**, and a player interface panel **2230**. The gaming display **2220** may include mechanical spinning reels (FIG. **23A**), a video display (FIGS. **23B** and **23C**), or a combination of both spinning reels and a video display (not shown). The gaming cabinet **2215** may also include a credit meter **2227** and a coin-in or bet meter **2228**. The credit meter **2227** may indicate the total number of credits remaining on the gaming device **2210** that are eligible to be wagered. In some embodiments, the credit meter **2227** may reflect a monetary unit, such as dollars. However, it is often preferable to have the credit meter **2227** reflect a number of ‘credits,’ rather than a monetary unit. The bet meter **2228** may indicate the amount of credits to be wagered on a particular game. Thus, for each game, the player transfers the amount that he or she wants to wager from the credit meter **2227** to the bet meter **2228**. In some embodiments, various other meters may be present, such as meters reflecting amounts won, amounts paid, or the like. In embodiments where the gaming display **2220** is a video monitor, the information indicated on the credit meters may be shown on the gaming display itself **2220** (FIG. **23B**).

[0126] The base portion **2213** may include a lighted panel **2214**, a coin return (not shown), and a gaming handle **2212** operable on a partially rotating pivot joint **2211**. The game handle **2212** is traditionally included on mechanical spinning-reel games, where the handle may be pulled toward a

player to initiate the spinning of reels **2222** after placement of a wager. The top box **2218** may include a lighted panel **2217**, a video display (such as an LCD monitor), a mechanical bonus device (not shown), and a candle light indicator **2219**. The player interface panel **2230** may include various devices so that a player can interact with the gaming device **10**.

[0127] The player interface panel **2230** may include one or more game buttons **2232** that can be actuated by the player to cause the gaming device **2210** to perform a specific action. For example, some of the game buttons **2232** may cause the gaming device **2210** to bet a credit to be wagered during the next game, change the number of lines being played on a multi-line game, cash out the credits remaining on the gaming device (as indicated on the credit meter **2227**), or request assistance from casino personnel, such as by lighting the candle **2219**. In addition, the player interface panel **2230** may include one or more game actuating buttons **2233**. The game actuating buttons **2233** may initiate a game with a pre-specified amount of credits. On some gaming devices **2210** a “Max Bet” game actuating button **2233** may be included that places the maximum credit wager on a game and initiates the game. The player interface panel **2230** may further include a bill acceptor **2237** and a ticket printer **2238**. The bill acceptor **2237** may accept and validate paper money or previously printed tickets with a credit balance. The ticket printer **2238** may print out tickets reflecting the balance of the credits that remain on the gaming device **2210** when a player cashes out by pressing one of the game buttons **2232** programmed to cause a ‘cashout.’ These tickets may be inserted into other gaming machines or redeemed at a cashier station or kiosk for cash.

[0128] The gaming device **2210** may also include one or more speakers **2226** to transmit auditory information or sounds to the player. The auditory information may include specific sounds associated with particular events that occur during game play on the gaming device **2210**. For example, a particularly festive sound may be played during a large win or when a bonus is triggered. The speakers **2226** may also transmit “attract” sounds to entice nearby players when the game is not currently being played.

[0129] The gaming device **2210** may further include a secondary display **2225**. In addition, a player's mobile computing device, typically a cellular phone, may supplement or replace secondary display **2225**. The phone may have an app that the player can download from an app store or a website that is compatible with the communications methods used in the gaming system. Alternatively, a web browser on the player's phone may operate as the secondary display. The phone may interface with the gaming system network via a cellular network, a WiFi network, or any other suitable wireless communications means. The secondary display **2225** may be a vacuum fluorescent display (VFD), a liquid crystal display (LCD), a cathode ray tube (CRT), a plasma screen, or the like. The secondary display **2225** may show any combination of primary game information and ancillary information to the player. For example, the secondary display **2225** may show player tracking information, secondary bonus information, advertisements, or player selectable game options.

[0130] The gaming device **2210** may include a separate information window (not shown) dedicated to supplying any combination of information related to primary game play, secondary bonus information, player tracking information, secondary bonus information, advertisements or player selectable game options. This window may be fixed in size and location or may have its size and location vary temporally as communication needs change. One example of such a resizable window is International Game Technology's “service window.” Another example is Las Vegas Gaming Incorporated's retrofit technology which allows information to be placed over areas of the game or the secondary display screen at various times and in various situations.

[0131] The gaming device **2210** includes a microprocessor **2240** that controls operation of the gaming device **2210**. If the gaming device **2210** is a standalone gaming device, the microprocessor **2240** may control virtually all of the operations of the gaming devices and attached equipment, such as operating game logic stored in memory (not shown) as firmware, controlling the display

2220 to represent the outcome of a game, communicating with the other peripheral devices (such as the bill acceptor **2237**), and orchestrating the lighting and sound emanating from the gaming device **2210**. In other embodiments where the gaming device **2210** is coupled to a network **2250**, as described below, the microprocessor **2240** may have different tasks depending on the setup and function of the gaming device. For example, the microprocessor **2240** may be responsible for running the base game of the gaming device and executing instructions received over the network **2250** from a bonus server or player tracking server. In a server-based gaming setup, the microprocessor **2240** may act as a terminal to execute instructions from a remote server that is running game play on the gaming device.

[0132] The microprocessor **2240** may be coupled to a machine communication interface (MCI) **4222** that connects the gaming device **2210** to a gaming network **2250**. The MCI **2242** may be coupled to the microprocessor **2240** through a serial connection, a parallel connection, an optical connection, or in some cases a wireless connection. The gaming device **2210** may include memory **2241** (MEM), such as a random access memory (RAM), coupled to the microprocessor **2240** and which can be used to store gaming information, such as storing total coin-in statistics about a present or past gaming session, which can be communicated to a remote server or database through the MCI **2242**. The MCI **2242** may also facilitate communication between the network **2250** and the secondary display **2225** or a player tracking unit **2245** housed in the gaming cabinet **2215**.

[0133] The player tracking unit **2245** may include an identification device **2246** and one or more buttons **2247** associated with the player tracking unit **2245**. The identification device **2246** serves to identify a player, by, for example, reading a player-tracking device, such as a player tracking card that is issued by the casino to individual players who choose to have such a card. The identification device **2246** may instead, or additionally, identify players through other methods. Player tracking systems using player tracking cards and card readers **2246** are known in the art. Briefly summarizing such a system, a player registers with the casino prior to commencing gaming. The casino issues a unique player-tracking card to the player and opens a corresponding player account that is stored on a server or host computer, described below with reference to FIG. **23**. The player account may include the player's name and mailing address and other information of interest to the casino in connection with marketing efforts. Prior to playing one of the gaming devices in the casino, the player inserts the player tracking card into the identification device **2246** thus permitting the casino to track player activity, such as amounts wagered, credits won, and rate of play.

[0134] To induce the player to use the card and be an identified player, the casino may award each player points proportional to the money or credits wagered by the player. Players typically accrue points at a rate related to the amount wagered, although other factors may cause the casino to award the player various amounts. The points may be displayed on the secondary display **2225** or using other methods. In conventional player tracking systems, the player may take his or her card to a special desk in the casino where a casino employee scans the card to determine how many accrued points are in the player's account. The player may redeem points for selected merchandise, meals in casino restaurants, or the like, which each have assigned point values. In some player tracking systems, the player may use the secondary display **2225** to access their player tracking account, such as to check a total number of points, redeem points for various services, make changes to their account, or download promotional credits to the gaming device **2210**. In other embodiments, the identification device **2246** may read other identifying cards (such as driver licenses, credit cards, etc.) to identify a player and match them to a corresponding player tracking account. Although FIG. **22A** shows the player tracking unit **2245** with a card reader as the identification device **2246**, other embodiments may include a player tracking unit **2245** with a biometric scanner, PIN code acceptor, or other methods of identifying a player to pair the player with their player tracking account.

[0135] During typical play on a gaming device **2210**, a player plays a game by placing a wager and then initiating a gaming session. The player may initially insert monetary bills or previously printed tickets with a credit value into the bill acceptor **2237**. The player may also put coins into a coin

acceptor (not shown) or a credit, debit or casino account card into a card reader/authorizer (not shown). In other embodiments, stored player points or special ‘bonus points’ awarded to the player or accumulated and/or stored in a player account may be able to be substituted at or transferred to the gaming device **2210** for credits or other value. For example, a player may convert stored loyalty points to credits or transfer funds from his bank account, credit card, casino account or other source of funding. The selected source of funding may be selected by the player at time of transfer, determined by the casino at the time of transfer or occur automatically according to a predefined selection process. One of skill in the art will readily see that the methods and systems described herein are useful with all gambling devices, regardless of the manner in which wager value-input is accomplished.

[0136] The credit meter **2227** may be configured to display the numeric credit value of the money or other value inserted, transferred, or stored dependent on the denomination of the gaming device **2210**. That is, if the gaming device **2210** is a nickel slot machine and a \$20 bill inserted into the bill acceptor **2237**, the credit meter will reflect 400 credits or one credit for each nickel of the inserted twenty dollars. For gaming devices **2210** that support multiple denominations, the credit meter **2227** will reflect the amount of credits relative to the denomination selected. Thus, in the above example, if a penny denomination is selected after the \$20 is inserted the credit meter will change from 400 credits to 2000 credits.

[0137] A wager may be placed by pushing one or more of the game buttons **2232**, which may be reflected on the bet meter **2228**. That is, the player can generally depress a “bet one” button (one of the buttons on the player interface panel **2230**, such as **2232**), which transfers one credit from the credit meter **2227** to the bet meter **2228**. Each time the button **2232** is depressed an additional single credit transfers to the bet meter **2228** up to a maximum bet that can be placed on a single play of the electronic gaming device **2210**. The gaming session may be initiated by pulling the gaming handle **2212** or depressing the spin button **2233**. On some gaming devices **2210**, a “max bet” button (another one of the buttons **2232** on the player interface panel **2230**) may be depressed to wager the maximum number of credits supported by the gaming device **2210** and initiate a gaming session.

[0138] If the gaming session does not result in any winning combination, the process of placing a wager may be repeated by the player. Alternatively, the player may cash out any remaining credits on the credit meter **2227** by depressing the “cash-out” button (another button **2232** on the player interface panel **2230**), which causes the credits on the credit meter **2227** to be paid out in the form of a ticket through the ticket printer **2238**, or may be paid out in the form of returning coins from a coin hopper (not shown) to a coin return tray.

[0139] If instead a winning combination (win) appears on the display **2220**, the award corresponding to the winning combination is immediately applied to the credit meter **2227**. For example, if the gaming device **2210** is a slot machine, a winning combination of symbols **2223** may land on a played payline on reels **2222**. If any bonus games are initiated, the gaming device **2210** may enter into a bonus mode or simply award the player with a bonus amount of credits that are applied to the credit meter **2227**.

[0140] FIGS. **23A** to **23C** illustrate exemplary types of gaming devices. FIG. **23A** illustrates an example spinning-reel gaming machine **2310A**, FIG. **23B** illustrates an example video slot machine **2310B**, and FIG. **23C** illustrates an example video poker machine **2310C**.

[0141] Referring to FIG. **23A**, a spinning-reel gaming machine **2310A** includes a gaming display **2320A** having a plurality of mechanical spinning reels **2322A**. Typically, spinning-reel gaming machines **2310A** have three to five spinning reels **2322A**. Each of the spinning reels **2322A** has multiple symbols **2323A** that may be separated by blank areas on the spinning reels **2322A**, although the presence of blank areas typically depends on the number of reels **2322A** present in the gaming device **2310A** and the number of different symbols **2323A** that may appear on the spinning reels **2322A**. Each of the symbols **2322A** or blank areas makes up a “stop” on the spinning reel

2322A where the reel **2322A** comes to rest after a spin. Although the spinning reels **2322A** of various games **2310A** may have various numbers of stops, many conventional spinning-reel gaming devices **2310A** have reels **2322A** with twenty two stops.

[0142] During game play, the spinning reels **222A** may be controlled by stepper motors (not shown) under the direction of the microprocessor **2240** (FIG. **22A**). Thus, although the spinning-reel gaming device **2310A** has mechanical based spinning reels **2322A**, the movement of the reels themselves is electronically controlled to spin and stop. This electronic control is advantageous because it allows a virtual reel strip to be stored in the memory **2341** of the gaming device **2310A**, where various “virtual stops” are mapped to each physical stop on the physical reel **2322A**. This mapping allows the gaming device **2310A** to establish greater awards and bonuses available to the player because of the increased number of possible combinations afforded by the virtual reel strips.

[0143] A gaming session on a spinning reel slot machine **2310A** typically includes the player pressing the “bet-one” button (one of the game buttons **2332A**) to wager a desired number of credits followed by pulling the gaming handle **2312** (FIGS. **22A**, **22B**) or pressing the spin button **2333A** to spin the reels **2322A**. Alternatively, the player may simply press the “max-bet” button (another one of the game buttons **2332A**) to both wager the maximum number of credits permitted and initiate the spinning of the reels **22A**. The spinning reels **2322A** may all stop at the same time or may individually stop one after another (typically from left to right) to build player anticipation. Because the display **2320A** usually cannot be physically modified, some spinning reel slot machines **2310A** include an electronic display screen in the top box **2318** (FIG. **22B**), a mechanical bonus mechanism in the top box **2318**, or a secondary display **2325** (FIG. **22A**) to execute a bonus.

[0144] Referring to FIG. **23B**, a video gaming machine **2310B** may include a video display **2320B** to display virtual spinning reels **2322B** and various other gaming information **2321B**. The video display **2320B** may be a CRT, LCD, plasma screen, or the like. It is usually preferable that the video display **2320B** be a touchscreen to accept player input. A number of symbols **2323A** appear on each of the virtual spinning reels **2322B**. Although FIG. **23B** shows five virtual spinning reels **2322B**, the flexibility of the video display **2320B** allows for various reel **2322B** and game configurations. For example, some video slot games **10B** spin reels for each individual symbol position (or stop) that appears on the video display **2320B**. That is, each symbol position on the screen is independent of every other position during the gaming sessions. In these types of games, very large numbers of pay lines or multiple super scatter pays can be utilized since similar symbols could appear at every symbol position on the video display **2320B**. On the other hand, other video slot games **2310B** more closely resemble the mechanical spinning reel games where symbols that are vertically adjacent to each other are part of the same continuous virtual spinning reel **2322B**.

[0145] Because the virtual spinning reels **2322B**, by virtue of being computer implemented, can have almost any number of stops on a reel strip, it is much easier to have a greater variety of displayed outcomes as compared to spinning-reel slot machines **2310A** (FIG. **23A**) that have a fixed number of physical stops on each spinning reel **2322A**.

[0146] With the possible increases in reel **2322B** numbers and configurations over the mechanical gaming device **2310A**, video gaming devices **2310B** often have multiple paylines **2324** that may be played. By having more paylines **2324** available to play, the player may be more likely to have a winning combination when the reels **2322B** stop and the gaming session ends. However, since the player typically must wager at least a minimum number of credits to enable each payline **2324** to be eligible for winning, the overall odds of winning are not much different, if at all, than if the player is wagering only on a single payline. For example, in a five line game, the player may bet one credit per payline **2324** and be eligible for winning symbol combinations that appear on any of the five played paylines **2324**. This gives a total of five credits wagered and five possible winning paylines **2324**. If, on the other hand, the player only wagers one credit on one payline **2324**, but plays five gaming sessions, the odds of winning would be identical as above: five credits wagered and five possible winning paylines **2324**.

[0147] Because the video display **2320B** can easily modify the image output by the video display **2320B**, bonuses, such as second screen bonuses are relatively easy to award on the video slot game **2310B**. That is, if a bonus is triggered during game play, the video display **2320B** may simply store the resulting screen shot in memory and display a bonus sequence on the video display **2320B**. After the bonus sequence is completed, the video display **2320B** may then retrieve the previous screen shot and information from memory, and re-display that image.

[0148] Also, as mentioned above, the video display **2320B** may allow various other game information **2321B** to be displayed. For example, as shown in FIG. **23B**, banner information may be displayed above the spinning reels **2322B** to inform the player, perhaps, which symbol combination is needed to trigger a bonus. Also, instead of providing a separate credit meter **2327** (FIG. **22A**) and bet meter **2328**, the same information can instead be displayed on the video display **2320B**. In addition, “soft buttons” **2329B** such as a “spin” button or “help/see pays” button may be built using the touch screen video display **2320B**. Such customization and ease of changing the image shown on the display **2320B** adds to the flexibility of the game **2310B**.

[0149] Even with the improved flexibility afforded by the video display **2320B**, several physical buttons **2332B** and **2333B** are usually provided on video slot machines **2310B**. These buttons may include game buttons **2332B** that allow a player to choose the number of paylines **2324** he or she would like to play and the number of credits wagered on each payline **2324**. In addition, a max bet button (one of the game buttons **2332B**) allows a player to place a maximum credit wager on the maximum number of available paylines **2324** and initiate a gaming session. A repeat bet or spin button **2333B** may also be used to initiate each gaming session when the max bet button is not used.

[0150] Referring to FIG. **23C**, a video poker gaming device **2310C** may include a video display **2320C** that is physically similar to the video display **2320B** shown in FIG. **23B**. The video display **2320C** may show a poker hand of five cards **2323C** and various other player information **2321C** including a payable for various winning hands, as well as a plurality of player selectable soft buttons **2329C**. The video display **2320C** may present a poker hand of five cards **23C** and various other player information **2321C** including a number of player selectable soft (touch-screen) buttons **2329C** and a payable for various winning hands. Although the embodiment illustrated in FIG. **23C** shows only one hand of poker on the video display **2320C**, various other video poker machines **2310C** may show several poker hands (multi-hand poker). Typically, video poker machines **2310C** play “draw” poker in which a player is dealt a hand of five cards, has the opportunity to hold any combination of those five cards, and then draws new cards to replace the discarded ones. All pays are usually given for winning combinations resulting from the final hand, although some video poker games **2310C** may give bonus credits for certain combinations received on the first hand before the draw. In the example shown in FIG. **23C** a player has been dealt two aces, a three, a six, and a nine. The video poker game **2310C** may provide a bonus or payout for the player having been dealt the pair of aces, even before the player decides what to discard in the draw. Since pairs, three of a kind, etc. are typically needed for wins, a player would likely hold the two aces that have been dealt and draw three cards to replace the three, six, and nine in the hope of receiving additional aces or other cards leading to a winning combination with a higher award amount. After the draw and revealing of the final hand, the video poker game **2310C** typically awards any credits won to the credit meter.

[0151] The player selectable soft buttons **2329C** appearing on the screen respectively correspond to each card on the video display **2320C**. These soft buttons **2329C** allow players to select specific cards on the video display **2320C** such that the card corresponding to the selected soft button is “held” before the draw. Typically, video poker machines **2310C** also include physical game buttons **2332C** that correspond to the cards in the hand and may be selected to hold a corresponding card. A deal/draw button **2333C** may also be included to initiate a gaming session after credits have been wagered (with a bet button **2332C**, for example) and to draw any cards not held after the first hand

is displayed.

[0152] Although examples of a spinning reel slot machine **2310A**, a video slot machine **10B**, and a video poker machine **2310C** have been illustrated in FIGS. **23A-23C**, gaming machines and various other types of gaming devices known in the art are contemplated.

[0153] FIG. **24** is a block diagram illustrating networked gaming devices. Referring to FIG. **24**, multiple electronic gaming devices (EGMs) **2470**, **2471**, **2472**, **2473**, **2474**, and **2475** may be coupled to one another and coupled to a remote server **2480** through a network **2450**. For ease of understanding, gaming devices or EGMs **2470**, **2471**, **2472**, **2473**, **2474**, and **2475** are generically referred to as EGMs **2470-2475**. The term EGMs **2470-2475**, however, may refer to any combination of one or more of EGMs **2470**, **2471**, **2472**, **2473**, **2474**, and **2475**. Additionally, the gaming server **2480** may be coupled to one or more gaming databases **2490**. These gaming network **2450** connections may allow multiple gaming devices **2470-2475** to remain in communication with one another during particular gaming modes such as tournament play or remote head-to-head play. Although some of the gaming devices **2470-2475** coupled on the gaming network **2450** may resemble the gaming devices **2210**, **2310A**, **2310B**, and **2310C** shown in FIGS. **22A-22B** and **23A-23C**, other coupled gaming devices **2470-2475** may include differently configured gaming devices. For example, the gaming devices **2470-2475** may include traditional slot machines **2475** directly coupled to the network **2450**, banks of gaming devices **2470** coupled to the network **2450**, banks of gaming devices **2470** coupled to the network **2450** through a bank controller **2460**, wireless handheld gaming machines **2472** and cell phones **2473** coupled to the gaming network **2450** through one or more wireless routers or antennas **2461**, personal computers **2474** coupled to the network **2450** through the internet **2462**, and banks of gaming devices **2471** coupled to the network through one or more optical connection lines **2464**. Additionally, some of the traditional gaming devices **2470**, **2471**, and **2475** may include electronic gaming tables, multi-station gaming devices, or electronic components operating in conjunction with non-gaming components, such as automatic card readers, chip readers, and chip counters, for example.

[0154] Gaming devices **2471** coupled over an optical line **2464** may be remote gaming devices in a different location or casino. The optical line **2464** may be coupled to the gaming network **2450** through an electronic to optical signal converter **2463** and may be coupled to the gaming devices **2471** through an optical to electronic signal converter **2465**. The banks of gaming devices **2470** coupled to the network **2450** may be coupled through a bank controller **2460** for compatibility purposes, for local organization and control, or for signal buffering purposes. The network **2450** may include serial or parallel signal transmission lines and carry data in accordance with data transfer protocols such as Ethernet transmission lines, Rs-232 lines, firewire lines, USB lines, or other communication protocols. Although not shown in FIG. **24**, substantially the entire network **2450** may be made of fiber optic lines or may be a wireless network utilizing a wireless protocol such as IEEE 802.11 a, b, g, or n, Zigbee, RF protocols, optical transmission, near-field transmission, or the like.

[0155] As mentioned above, each gaming device **2470-2475** may have an individual processor **2440** (FIG. **22A**) and memory **2441** to run and control game play on the gaming device **2470-2475**, or some of the gaming devices **2470-2475** may be terminals that are run by a remote server **2480** in a server based gaming environment. Server based gaming environments may be advantageous to casinos by allowing fast downloading of particular game types or themes based on casino preference or player selection. Additionally, tournament based games, linked games, and certain game types, such as BINGO or keno may benefit from at least some server **80** based control.

[0156] Thus, in some embodiments, the network **2450**, server **2480**, and database **2490** may be dedicated to communications regarding specific game or tournament play. In other embodiments, however, the network **2450**, server **2480**, and database **2490** may be part of a player tracking network. For player tracking capabilities, when a player inserts a player tracking card in the card reader **2446** (FIG. **22A**), the player tracking unit **2245** sends player identification information

obtained on the card reader **2246** through the MCI **2242** over the network **2250** to the player tracking server **2480**, where the player identification information is compared to player information records in the player database **2490** to provide the player with information regarding their player account or other features at the gaming device **2210** where the player is wagering. Additionally, multiple databases **2490** and/or servers **2480** may be present and coupled to one or more networks **2250** to provide a variety of gaming services, such as both game/tournament data and player tracking data.

[0157] The various systems described with reference to FIGS. **22A-24** can be used in a number of ways. For instance, the systems can be used to track data about various players. The tracked data can be used by the casino to provide additional benefits to players, such as extra bonuses or extra benefits such as bonus games and other benefits as described above. These added benefits further entice the players to play at the casino that provides the benefits.

[0158] FIG. **25** shows an example method **2500**. The method **2500** may be carried out via any one or more devices as described herein. At **2510**, a user input may be received. The user input may be received by an electronic gaming device. The electronic gaming machine may be a casino gaming machine (e.g., a slot machine device, a virtual slot machine device, a virtual gaming table device, a virtual card gaming device, combinations thereof, and the like). The electronic gaming machine may be configured to store one or more credits. The electronic gaming machine may be configured to output a visual indicator associated with the one or more credits (e.g., a credit meter). The user input may comprise a designated amount of credits of the one or more credits.

[0159] At **2520**, one or more physical tokens may be generated. The one or more physical tokens may be generated by the electronic gaming machine. Generating the one or more physical tokens may comprise outputting one or more tickets (e.g., printing one or more tickets), one or more chips (e.g., via a chip dispenser), combinations thereof, and the like. Generating the one or more physical tokens comprising the enhanced value of the designated amount of credits may comprise transferring the designated amount of credits from a credit meter on the electronic gaming machine to the one or more physical tokens. Generating the one or more physical tokens may comprise printing the enhanced value on a ticket.

[0160] The one or more physical tokens may comprise an enhanced value of the designated amount of credits. The enhanced value may comprise a modification to, restriction associated with, rule associated with, or time parameter associated with the designated amount. The one or more physical tokens may comprise one or more indicia indicating the enhanced value. For example, the designated amount may be 10 credits, but the enhanced value is 15 credits if a player uses the one or more physical tokens at a particular time, at a particular machine, at a particular sportsbook, or the like. For example, if the player returns to the casino in one month, the designated amount may increase from 10 credits to 15 credits. For example, the enhanced value may be determined based on player information such as player account information (e.g., whether the player is a member of loyalty club), historical game play data, wagering patterns, one or more indicators associated with problem gambling, combinations thereof, or the like.

[0161] The designated amount may be associated with a trigger event. The trigger event may be user determined. For example, a user may set a loss-stop limit. The loss-stop limit may be configured to prevent the player from losing all of the credits in the credit meter. For example, upon the credit meter reaching a certain number of credits, the user may be prompted with an option to receive an enhanced value ticket if the player cashes out a designated amount of credits.

[0162] The enhanced value of the designated amount of credits may comprise one or more of: eligibility for a multiple jackpot on a future game played, eligibility for a bonus on a future game played, a drawing entry, a tournament entry, or an increase in credits over the transferred amount of credits.

[0163] The trigger event may be determined by the system. For example, the electronic gaming machine or another device may be configured to determine problem gambling behavior. For

example, the player may be associated with a player account. The player account may contain financial information about the player (e.g., salary, total asserts, or other information). The system may be configured to offer the player an option to designate an amount of credits upon the credit meter reaching some threshold determined based on the financial information about the player. For example, if a wager amount is equal to or greater than a certain percentage of the player's salary, the system may offer the player to designate an amount and thereby receive the one or more enhanced value tokens. Similarly, the player may be enrolled in a self-exclusion program. The player's enrollment in the self-exclusion program may be included in the player account associated with the player.

[0164] At **2530**, the player may be permitted to use the one or more physical tokens for additional game play. For example, the player may be permitted to use the one or more physical tokens based on the enhanced value of the designated amount of credits. For example, the player could buy designate 10 credits and the enhanced value may be 20 credits conditioned on the player playing the one or more physical tokens at a specific game type within a defined period of time. Permitting the player to use the enhanced value for additional game play may comprise conditioning use of the generated one or more physical tokens on at least one of playing a specified game, playing at a specified time, or placing a sports bet. For example, the electronic gaming machine may display a message to the player saying "Your luck is running low on this machine. We just installed a 'Green Martian' machine that we think you'll like. Cash out 10 credits now and we'll give you a ticket worth 20 credits if you use it at a Green Martian machine today."

[0165] The method may comprise determining a payback percentage variance associated with the one or more electronic gaming machines. For example, a payback percentage associated with a first electronic gaming machine may be determined. The first electronic gaming machine may be the machine the player is currently playing. A variance from a payback percentage associated with the first electronic gaming machine may be determined. An opportunity for an enhanced value token may be presented to the player based on the variance from the payback percentage associated with the first electronic gaming machine. A variance from a payback percentage associated with a second electronic gaming machine may be determined. The second electronic gaming machine may not be the electronic gaming machine currently being played by the player. The enhanced value token may be configured to be playable at the second electronic gaming machine, thereby incentivizing the player to stop playing the first electronic gaming machine and to continue play on the second electronic gaming machine. For example, it may be determined that a large jackpot was just paid out by the second electronic gaming machine, thus creating the variance from the payback percentage associated with the second electronic gaming machine. It may be determined that play has ceased at the second electronic gaming machine because, for example, the player who won the jackpot on the second electronic gaming machine has withdrawn his player identifier card, or perhaps his mobile device is no longer in proximity to the second electronic gaming machine. The player at the first electronic gaming machine may be presented with an option for an enhanced value token configured to be played at the second electronic gaming machine.

[0166] The method may comprise multiplying one or more outcome values (e.g., points, credits, or any other type of value). For example, the method may comprise multiplying the outcome values for a predetermined number of future plays.

[0167] FIG. **26** shows an example method **2600**. The method may be carried out via any one or more devices as described herein. For example, at **2610**, a user input may be received. The user input may be received by an electronic casino gaming machine. The user input may be configured to cause the electronic casino gaming machine to transfer an amount of credits and/or other value to a physical token. The user input may be received via one or more of: a touch screen associated with the gaming system or a mobile computing device associated with the gaming system. For example, the physical token may comprise a ticket, a chip, a card, combinations thereof, and the like.

Transferring the amount of credits to the physical token may comprise executing an electronic

funds transfer. For example, the electronic funds transfer may be configured to transfer funds from a player account to casino account. For example, the casino account may comprise a casino agent account associated with one or more casino agents.

[0168] At **2620**, the physical token may be generated. Generating the physical token may comprise printing or otherwise outputting a ticket, a chip, or a card. For example, the electronic casino gaming machine may comprise a ticket printer, token dispenser, or other similar mechanism configured to output the physical token. The physical token may be generated based on the user input. The physical token may be generated based on the amount of credits indicated by the user input.

[0169] At **2630**, a credit pooling parameter may be determined. The credit pooling parameter may be determined based on the user input. The credit pooling parameter may be configured to indicate whether the one or more credits or other value should be added to a pooled account (e.g., a pooled tip account associated with a plurality of casino agents) or an individual account (e.g., an individual tip account associated with a single casino agent). The credit pooling parameter may be associated with the electronic casino gaming machine. For example, all electronic casino gaming machines in a first part of a casino may be associated with a credit pooling parameter configured to pool one or more credits. Similarly, all electronic casino gaming machines in a second part of the casino may be associated with a credit pooling parameter configured to direct the amount of credits to a credit account associated with a single casino agent (e.g., a dealer, a particular server).

[0170] At **2640**, the amount of credits may be stored. For example, the amount of credits may be stored based on the credit pooling parameter. The amount of credits may be stored in an electronic account comprising one or more second amounts of credits (e.g., one or more second amounts of credits previously contributed by other players).

[0171] The method may comprise determining a payback percentage variance associated with the one or more electronic gaming machines. For example, a payback percentage associated with a first electronic gaming machine may be determined. The first electronic gaming machine may be the machine the player is currently playing. A variance from a payback percentage associated with the first electronic gaming machine may be determined. An opportunity for an enhanced value token may be presented to the player based on the variance from the payback percentage associated with the first electronic gaming machine. A variance from a payback percentage associated with a second electronic gaming machine may be determined. The second electronic gaming machine may not be the electronic gaming machine currently being played by the player. The enhanced value token may be configured to be playable at the second electronic gaming machine, thereby incentivizing the player to stop playing the first electronic gaming machine and to continue play on the second electronic gaming machine. For example, it may be determined that a large jackpot was just paid out by the second electronic gaming machine, thus creating the variance from the payback percentage associated with the second electronic gaming machine. It may be determined that play has ceased at the second electronic gaming machine because, for example, the player who won the jackpot on the second electronic gaming machine has withdrawn his player identifier card, or perhaps his mobile device is no longer in proximity to the second electronic gaming machine. The player at the first electronic gaming machine may be presented with an option for an enhanced value token configured to be played at the second electronic gaming machine. The method may comprise multiplying one or more outcome values (e.g., points, credits, or any other type of value). For example, the method may comprise multiplying the outcome values for a predetermined number of future plays.

[0172] The methods and systems may be implemented on a computer **2701** as shown in FIG. 27 and described below. The methods and systems described may utilize one or more computers to perform one or more functions in one or more locations. FIG. 27 is a block diagram of an operating environment for performing the present methods. This operating environment is a single configuration of many possible configurations of an operating environment, and it is not intended

to suggest any limitation as to the scope of use or functionality of operating environment architecture. Neither should the operating environment be interpreted as having any dependency or requirement relating to any one or combination of components shown in the operating environment.

[0173] The present methods and systems may be operational with numerous other general purpose or special purpose computing system environments or configurations. Well-known computing systems, environments, and/or configurations that may be suitable for use with the systems and methods may be, but are not limited to, personal computers, server computers, laptop devices, and multiprocessor systems. Additional computing systems, environments, and/or configurations are set top boxes, programmable consumer electronics, network PCs, minicomputers, mainframe computers, distributed computing environments that are composed of any of the above systems or devices, and the like.

[0174] The processing of the present methods and systems may be performed by software components. The described systems and methods may be described in the general context of computer-executable instructions, such as program modules, being executed by one or more computers or other devices. Generally, program modules are composed of computer code, routines, programs, objects, components, data structures, etc. that perform particular tasks or implement particular abstract data types. The described methods may also be practiced in grid-based and distributed computing environments where tasks are performed by remote processing devices that are linked through a communications network. In a distributed computing environment, program modules may be located in both local and remote computer storage media including memory storage devices.

[0175] Further, one skilled in the art will appreciate that the systems and methods described herein may be implemented via a general-purpose computing device in the form of a computer **2701**. The components of the computer **2701** may be, but are not limited to, one or more processors **2703**, a system memory **2713**, and a system bus **2713** that couples various system components including the one or more processors **2703** to the system memory **2713**. The system may utilize parallel computing.

[0176] The system bus **2713** represents one or more of several possible types of bus structures, including a memory bus or memory controller, a peripheral bus, an accelerated graphics port, or local bus using any of a variety of bus architectures. Such architectures may be an Industry Standard Architecture (ISA) bus, a Micro Channel Architecture (MCA) bus, an Enhanced ISA (EISA) bus, a Video Electronics Standards Association (VESA) local bus, an Accelerated Graphics Port (AGP) bus, and a Peripheral Component Interconnects (PCI), a PCI-Express bus, a Personal Computer Memory Card Industry Association (PCMCIA), Universal Serial Bus (USB) and the like. The bus **2713**, and all buses specified in this description may also be implemented over a wired or wireless network connection and each of the subsystems, including the one or more processors **2703**, a mass storage device **2704**, an operating system **2705**, wagering software **2706**, wagering data **2707**, a network adapter **2708**, the system memory **2713**, an Input/Output Interface **2710**, a display adapter **2709**, a display device **2711**, and a human machine interface **2702**, may be contained within one or more remote computing devices **2714a,b,c** at physically separate locations, connected through buses of this form, in effect implementing a fully distributed system.

[0177] The computer **2701** is typically composed of a variety of computer readable media. Readable media may be any available media that is accessible by the computer **2701** and may be both volatile and non-volatile media, removable and non-removable media. The system memory **2713** may be computer readable media in the form of volatile memory, such as random access memory (RAM), and/or non-volatile memory, such as read only memory (ROM). The system memory **2713** is typically composed of data such as the wagering data **2707** and/or program modules such as the operating system **2705** and the object identification and wagering software **2706** that are immediately accessible to and/or are presently operated on by the one or more

processors **2703**.

[0178] The computer **2701** may also be composed of other removable/non-removable, volatile/non-volatile computer storage media. FIG. **27** shows a mass storage device **2704**, which may provide non-volatile storage of computer code, computer readable instructions, data structures, program modules, and other data for the computer **2701**. The mass storage device **2704** may be a hard disk, a removable magnetic disk, a removable optical disk, magnetic cassettes or other magnetic storage devices, flash memory cards, CD-ROM, digital versatile disks (DVD) or other optical storage, random access memories (RAM), read only memories (ROM), electrically erasable programmable read-only memory (EEPROM), and the like.

[0179] Optionally, any number of program modules may be stored on the mass storage device **2704**, such as the operating system **2705** and the wagering software **2706**. Each of the operating system **2705** and the wagering software **2706** (or some combination thereof) may be elements of the programming and the wagering software **2706**. The wagering data **2707** may also be stored on the mass storage device **2704**. The wagering data **2707** may be stored in any of one or more databases known in the art. Such databases are DB2®, Microsoft® Access, Microsoft® SQL Server, Oracle®, MySQL, PostgreSQL, and the like. The databases may be centralized or distributed across multiple systems.

[0180] The user may enter commands and information into the computer **2701** via an input device (not shown). Such input devices may be, but are not limited to, a keyboard, pointing device (e.g., a “mouse”), a microphone, a joystick, a scanner, tactile input devices such as gloves, and other body coverings, and the like. These and other input devices may be connected to the one or more processors **2703** via the human machine interface **2702** that is coupled to the system bus **2713**, but may be connected by other interface and bus structures, such as a parallel port, game port, an IEEE **2794** Port (also known as a Firewire port), a serial port, or a universal serial bus (USB).

[0181] The display device **2711** may also be connected to the system bus **2713** via an interface, such as the display adapter **2709**. It is contemplated that the computer **2701** may have more than one display adapter **2709** and the computer **2701** may have more than one display device **2711**. The display device **2711** may be a monitor, an LCD (Liquid Crystal Display), or a projector. In addition to the display device **2711**, other output peripheral devices may be components such as speakers (not shown) and a printer (not shown) which may be connected to the computer **2701** via the Input/Output Interface **2710**. Any step and/or result of the methods may be output in any form to an output device. Such output may be any form of visual representation, including, but not limited to, textual, graphical, animation, audio, tactile, and the like. The display device **2711** and computer **2701** may be part of one device, or separate devices.

[0182] The computer **2701** may operate in a networked environment using logical connections to one or more remote computing devices **2714a,b,c**. A remote computing device may be a personal computer, portable computer, smartphone, a server, a router, a network computer, a peer device or other common network node, and so on. Logical connections between the computer **2701** and a remote computing device **2714a,b,c** may be made via a network **2715**, such as a local area network (LAN) and/or a general wide area network (WAN). Such network connections may be through the network adapter **2708**. The network adapter **2708** may be implemented in both wired and wireless environments. Such networking environments are conventional and commonplace in dwellings, offices, enterprise-wide computer networks, intranets, and the Internet.

[0183] Application programs and other executable program components such as the operating system **2705** are shown herein as discrete blocks, although it is recognized that such programs and components reside at various times in different storage components of the computing device **2701**, and are executed by the one or more processors **2703** of the computer. An implementation of the wagering software **2706** may be stored on or sent across some form of computer readable media. Any of the described methods may be performed by computer readable instructions embodied on computer readable media. Computer readable media may be any available media that may be

accessed by a computer. Computer readable media may be “computer storage media” and “communications media.” “Computer storage media” may be composed of volatile and non-volatile, removable and non-removable media implemented in any methods or technology for storage of information such as computer readable instructions, data structures, program modules, or other data. Further, computer storage media may be, but are not limited to, RAM, ROM, EEPROM, flash memory or other memory technology, CD-ROM, digital versatile disks (DVD) or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium which may be used to store the desired information and which may be accessed by a computer.

[0184] The methods and systems may employ Artificial Intelligence techniques such as machine learning and iterative learning. Such techniques include, but are not limited to, expert systems, case based reasoning, Bayesian networks, behavior based AI, neural networks, fuzzy systems, evolutionary computation (e.g. genetic algorithms), swarm intelligence (e.g. ant algorithms), and hybrid intelligent systems (e.g. Expert inference rules generated through a neural network or production rules from statistical learning).

[0185] Unless otherwise expressly stated, it is in no way intended that any method set forth herein be construed as requiring that its steps be performed in a specific order. Accordingly, where a method claim does not actually recite an order to be followed by its steps or it is not otherwise specifically stated in the claims or descriptions that the steps are to be limited to a specific order, it is in no way intended that an order be inferred, in any respect. This holds for any possible non-express basis for interpretation, including: matters of logic with respect to arrangement of steps or operational flow; plain meaning derived from grammatical organization or punctuation; and the number or type of configurations described in the specification.

[0186] It will be apparent to those skilled in the art that various modifications and variations may be made without departing from the scope or spirit. Other configurations will be apparent to those skilled in the art from consideration of the specification and practice described herein. It is intended that the specification and methods and systems described therein be considered exemplary only, with a true scope and spirit being indicated by the following claims.

Claims

1. A method comprising: receiving, by an electronic gaming machine configured to store one or more credits, a user input comprising a designated amount of credits of the one or more credits; based on receiving the user input, generating, based on the designated amount of credits, one or more physical tokens comprising an enhanced value of the designated amount of credits; and permitting, based on the enhanced value of the designated amount of credits, the player to use the one or more physical tokens for additional game play.
2. The method of claim 1, wherein the designated amount of credits is less than the one or more credits, the method further comprising multiplying, based on a number of future plays, one or more game outcome values.
3. The method of claim 1, wherein the enhanced value of the designated amount of credits comprises one or more of: eligibility for a multiple jackpot on a future game played, eligibility for a bonus on a future game played, a drawing entry, a tournament entry, or an increase in credits over the transferred amount of credits.
4. The method of claim 1, wherein generating the one or more physical tokens comprising the enhanced value of the designated amount of credits comprises transferring the designated amount of credits from a credit meter on the electronic gaming machine to the one or more physical tokens.
5. The method of claim 1, wherein the one or more physical tokens comprise indicia indicating the enhanced value.
6. The method of claim 1, wherein generating the one or more physical tokens comprises printing

the enhanced value on a ticket, and wherein permitting the player to use the enhanced value for additional game play comprises conditioning use of the generated one or more physical tokens on at least one of playing a specified game, playing at a specified time, or placing a sports bet.

7. The method of claim 1, wherein the user input is received by the electronic gaming machine via one or more of: a touch screen display or a mobile computing device.

8. A casino gaming system comprising: one or more electronic casino gaming machines; a mechanism configured to generate one or more physical tokens; at least one processor coupled to the mechanism; at least one non-transitory computer readable medium storing processor executable instructions thereon, which, when executed by the at least one processor, cause the at least one processor to: receive, by an electronic gaming machine of the one or more electronic casino gaming machines, a user input comprising a designated amount of credits of one or more credits, wherein the electronic casino gaming machine is configured to store the one or more credits, based on receiving the user input, generate, based on the designated amount of credits, one or more physical tokens comprising an enhanced value of the designated amount of credits; and permit, based on the enhanced value of the designated amount of credits, the player to use the physical token for additional game play.

9. The system of claim 8, wherein the designated amount of credits is less than the one or more credits, wherein the computing device is further configured to multiply, based on a number of future plays, one or more game outcome values.

10. The system of claim 8, wherein the enhanced value of the designated amount of credits comprises at least one of: eligibility for a multiple jackpot on a future game played, eligibility for a bonus on a future game played, a drawing entry, a tournament entry, or an increase in credits over the transferred amount of credits.

11. The system of claim 8, wherein the processor executable instructions that, when executed by the at least one processor, cause the at least one processor to generate the physical token comprising the enhanced value of the designated amount of credits, cause the at least one processor to transfer the designated amount of credits from a credit meter on the electronic gaming machine to the physical token.

12. The system of claim 8, wherein the physical token comprises indicia indicating the enhanced value.

13. The system of claim 8, wherein the processor executable instructions that, when executed by the at least one processor, cause the at least one processor to generate the physical token comprising the enhanced value of the designated amount of credits, further cause the at least one processor to print the enhanced value on a ticket, and wherein permitting the player to use the enhanced value for additional game play comprises conditioning use of the generated physical token on at least one of playing a specified game, playing at a specified time, or placing a sports bet.

14. The system of claim 8, wherein the user input is received by the electronic gaming machine via one or more of: a touch screen display or a mobile computing device.

15. A method comprising: receiving, by an electronic casino gaming machine, a user input configured to cause the electronic casino gaming machine to transfer an amount of credits to a physical token; generating, based on the user input and the amount of credits, the physical token, wherein the physical token is configured to represent the amount of credits; determining a credit pooling parameter associated with the electronic casino gaming machine; and storing, based on the credit pooling parameter associated with the electronic casino gaming machine, the amount of credits in an electronic account.

16. The method of claim 15, wherein the user input is received via one or more of: a touch screen associated with the gaming system or a mobile computing device associated with the gaming system, the method further comprising multiplying, based on a number of future plays, one or more game outcome values.

- 17.** The method of claim 15, wherein the mechanism is a ticket printer and wherein generating the physical token comprises printing a ticket and wherein the physical token comprises indicia indicating an amount of credits associated with the physical token.
- 18.** The method of claim 15, wherein the credit pooling parameter indicates whether the one or more credits should be pooled and wherein the credit account comprises one or more second amounts of credits.
- 19.** A casino gaming system comprising: one or more electronic casino gaming machines; a mechanism for generating a physical token; at least one processor coupled to the mechanism; at least one non-transitory computer readable medium that stores a plurality of instructions, the plurality of instructions, when executed by the at least one processor, causing the at least one processor to: receive, by an electronic casino gaming machine of the one or more electronic casino machines, a user input configured to cause the electronic casino gaming machine to transfer an amount of credits to a physical token; generate, based on the user input and the amount of credits, the physical token, wherein the physical token is configured to represent the amount of credits; determine a credit pooling parameter associated with the electronic casino gaming machine; and store, based on the credit pooling parameter associated with the electronic casino gaming machine, the amount of credits in an electronic account comprising one or more second amounts of credits.
- 20.** The system of claim 19, wherein the request is received via one or more of: a touch screen associated with the gaming system or a mobile computing device associated with the gaming system, wherein the computing device is further configured to multiply, based on a number of future plays, one or more game outcome values.
- 21.** The system of claim 19, wherein the mechanism is a ticket printer and wherein generating the physical token comprises printing a ticket.
- 22.** The system of claim 19, wherein the physical token comprises indicia indicating an amount of credits associated with the physical token.
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